

Class Notes: Basis of Classification

Animal Kingdom • PU-I Biology • Class Notes

Why Classification Matters

Over a million species of animals have been described so far, making classification essential. Basis of classification includes: levels of organisation, patterns of digestive/circulatory/reproductive systems, body symmetry, arrangement of cells in germ layers, nature of coelom, segmentation, and presence or absence of notochord.

Criteria Used in Animal Classification

levels of organisation: cellular level (e.g. Porifera), tissue level (e.g. Coelenterata), organ level (e.g. Platyhelminthes), and organ system level (e.g. Echinoderms and Chordates). Organ system pattern may be incomplete (one opening, e.g. Platyhelminthes) or complete (separate mouth and anus, with open or closed circulation).